

DIRECTED ENERGY WEAPONS: More On Less-Lethal Laser Weapons



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Under directed energy weapons, we have covered particle beam weapons, high-power microwaves and less-lethal laser weapons. Let us learn more about less-lethal laser weapons.

Emerging trends

Laser dazzlers have shown a lot of promise and potential for a variety of low-intensity conflict applications including close-quarter battle scenarios, border patrol and coastal surveillance, controlling unruly and violent crowds and mobs, countering asymmetric threats particularly encountered in naval environment, protection of critical infrastructure from terrorist attacks and so on and so forth.

Different variants of these devices are being developed to suit different application requirements. While on one hand, handheld portable devices are being offered to meet short-range requirements of paramilitary and law-enforcement agencies, tripod-mounted and vehicle-mounted versions equipped with electro-optic sighting support, target acquisition and tracking devices are also being developed to meet the requirements of emerging scenarios.

While short- and medium-range handheld and weapon-integrated versions of laser dazzlers are in widespread use for anti-terrorist and counter-insurgency applications, vehicle-mounted systems designed for operations against unruly and violent crowds are beginning to appear on the scene. Ship-borne medium- to long-range laser dazzlers mounted on stabilised platforms are also catching the attention of security forces to defeat asymmetric threats.

Another interesting development in the recent past has been a keen interest in long-range laser dazzler systems mounted on Gimbal platforms and equipped with electro-optic trackers and also integrated

with a network of radars to provide 24 x 7 protection to strategic assets from airspace violation by aerial platforms.

RGB laser dazzlers that emit an intense flashing pattern of red, green and blue lights at a frequency of a few kilohertz are also reportedly under development, though not much information is currently available about the existence of any commercial devices. These devices may be configured around a combination of semiconductor diode lasers and diode-pumped solid-state laser modules emitting red, green and blue wavelengths. Arrays of red, green and blue LEDs could also be used instead.

RGB laser dazzlers have the potential of being more effective as compared to green laser dazzlers, particularly if the targeted personnel use protective eye-wear. Also, these devices can cause nausea in the exposed personnel, which makes RGB dazzlers particularly effective in sea environment. One possible application could be against sea pirates.

In not too distant future, one would see the deployment of laser dazzlers with global coverage (Fig. 11 in the previous part of the article). These systems are proposed to use remotely-controlled membrane reflectors to receive the dazzling laser beam from the source station and guide it to intended target location.

High-power lasers for ordnance disposal

Disposal of unexploded ordnances (UXOs) including surface-laid mines, improvised explosive devices (IEDs), grenade shells, artillery/mortar rounds, cluster bombs and so on from safe stand-off ranges using a high-power laser beam is an emerging application of directed energy laser systems in homeland-security-related applications.